

Q. Transformer Encoder - Decoder workflow worksheet

Ans 1. Tokenize a sentence :

Def: Tokenization is the process of breaking a sentence into smaller units called tokens.

Ex:

Sentence: "Artificial intelligence is transforming education".

Tokens:

["Artificial", "intelligence", "is", "transforming", "education"]

Purpose:

- split text into manageable pieces -
- Helps the model understand the structure of sentence
- prepares text for further processing

2. Convert Tokens to Token IDs

Def: A Token ID is a unique numerical value assigned to each token using the model vocabulary.

Ex:

Token	ID
Artificial	2031
intelligence	4567
is	2003
transforming	10938
education	2495
.	1012

output :

[2031, 4567, 2003, 10938, 2495, 1012]

Purpose :

- Converts Words into numerical form

3. Embeddings:

Def : An embedding is a numerical vector that represent the meaning of word or token

Ex :

Artificial $\rightarrow [0.21, -0.45, 0.67, \dots]$

Intelligence $\rightarrow [0.14, 0.82, 0.31, \dots]$

Purpose :

Embedding convert token IDs into meaningful vectors before they are processed by Transformers

4. Label a transformer encoder :

Def :

The Transformer encoder reads the input sentence and generates contextual representations for every word. It learn how word relates to every other word

5. Identify Query (Q), Key (K), Value (V)

Definition : Every input embedding is transformed into three vectors.

- Query
- Key
- Value.

- Query : Represent what a word is searching for
- Key : Represent what information a word contain
- Value : Represent the actual information passed to the output.

6. Explain which words Recieve Higher Attention

Def : Attention determines which words are most important while understanding a sentence. Related words recieve higher attention scores.

Ex :

“ The student solved the difficult problem ”

student

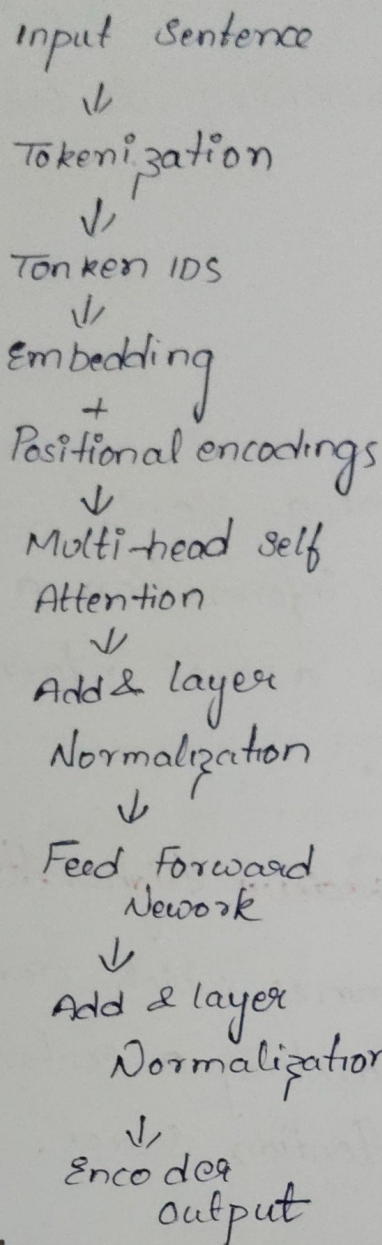
→ who perform action

difficult

→ describe problem.

Attention helps the transformer understand context and relationships between words.

Encoder components :



Purpose of each layer :

- **Embeddings** : convert words into vectors
- **positional encoding** : Adds word order information
- **Self-Attention** : Finds relationships between words
- **Feed Forward Network** : learns complex patterns

7. Compare transformer models:

Def: Different Transformer models are designed for NLP Tasks behind their Architecture.

	Architecture	Main use.
BERT	Encoder	Text classification
GPT	Decoder	Text generation
T5	Encoder-decoder	Translation, summarization
BART	Encoder-decoder	Text generation, correction.

8. Complete the Transformer pipeline:

Definition: The Transformer processes text through several stages before generating final output.

Pipeline:

Input sentence



Tokenization



Token IDs



Embeddings



Positional Encoding



Transformer encoder and decoder



linear layer

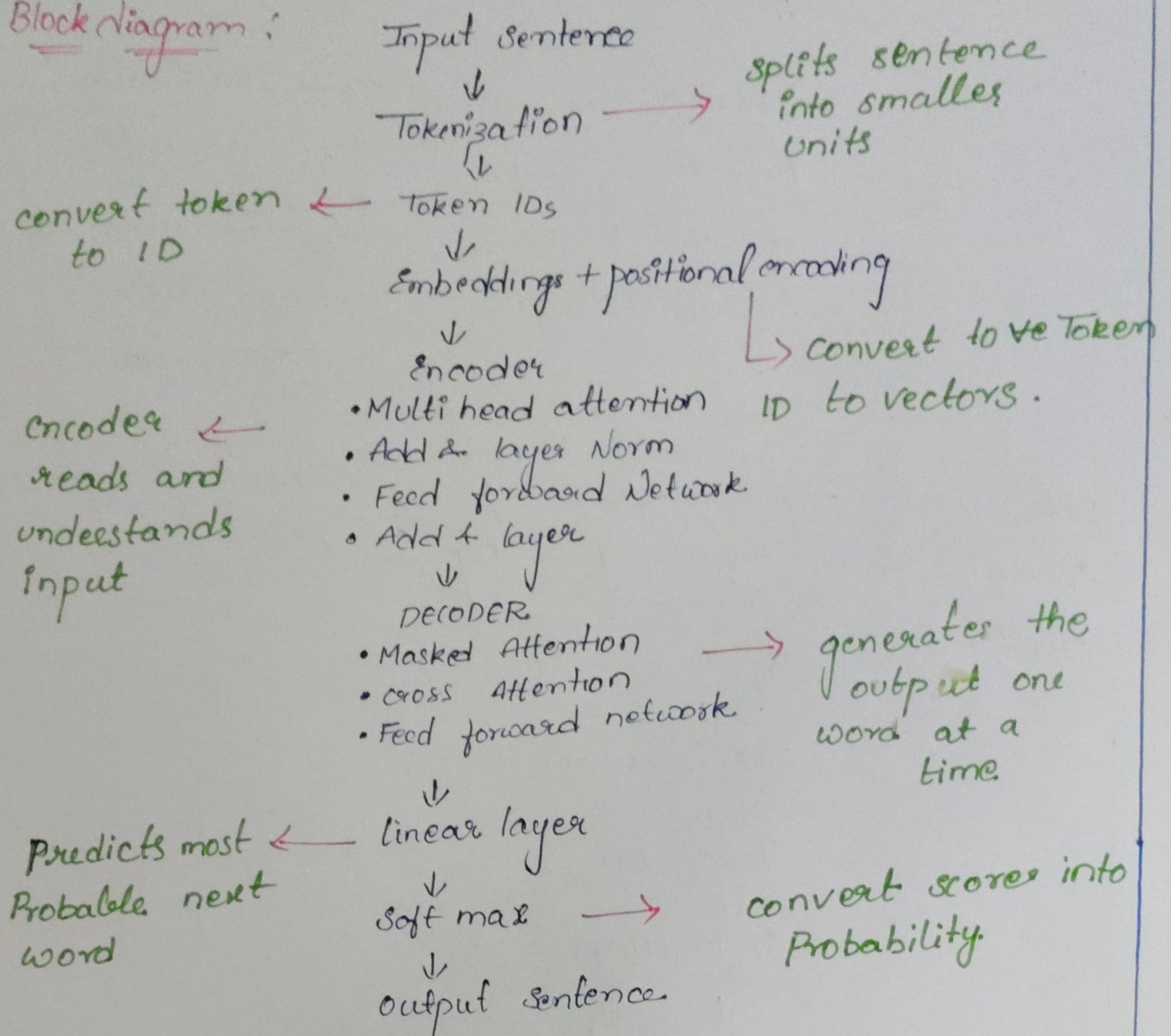


Soft max → output prediction

9. Transformer Block Diagram.

Definition: The transformer consist of Encoder that understand the input and a decoder that generates output.

Block diagram:



Conclusion:

The Transformer encoder-decoder architecture is foundation models for NLP such as BERT, GPT, T5. It process text by tokenizing input, converting the tokens to embeddings. Applying Attention using Q, K, V and finally generating output.